

BUDAPEST FERIHEGY, Hungary

WMO: 128390

Lat: 47.437N

Lon: 19.256E

Elev: 495

StdP: 14.43

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.8	15.8	6.5	7.8	14.9	11.8	10.2	18.3	28.9	43.8	25.3	41.2	3.3	320	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	91.6	71.6	88.1	70.0	85.3	68.9	73.7	87.0	72.0	84.2	70.2	81.6	9.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.7	111.1	81.2	68.0	104.7	77.9	66.3	98.7	75.6	37.7	86.9	35.9	84.0	34.5	81.6	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	20.2	17.2	DB	3.9	95.8	5.8	3.5	-0.3	98.4	-3.7	100.4	-7.0	102.4	-11.2	105.0
			WB	3.4	76.2	5.7	2.7	-0.7	78.2	-4.1	79.7	-7.3	81.2	-11.5	83.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	31.1	34.4	42.2	52.4	60.9	67.3	70.8	70.2	61.2	51.6	42.1	32.4	(6)
	DBStd	15.84	7.77	8.72	7.59	6.90	6.43	6.27	5.68	6.08	6.30	7.36	7.48	8.01	(7)
	HDD50	2207	585	439	256	53	4	0	0	0	1	71	253	545	(8)
	HDD65	5494	1050	857	707	380	159	46	14	21	147	416	688	1009	(9)
	CDD50	2749	0	2	14	125	341	520	646	626	338	122	15	0	(10)
	CDD65	560	0	0	0	2	31	116	195	181	34	1	0	0	(11)
	CDH74	6198	0	0	0	53	425	1299	2115	1935	354	17	0	0	(12)
	CDH80	2215	0	0	0	6	83	445	833	766	82	0	0	0	(13)
(14) Wind	WSAvg	7.0	6.7	7.2	8.1	8.0	7.3	7.1	7.4	6.3	6.6	6.4	6.9	6.6	(14)
(15) Precipitation	PrecAvg	21.5	1.3	1.2	1.2	1.5	2.3	2.5	2.5	2.2	1.9	1.5	2.0	1.5	(15)
	PrecMax	36.1	3.5	4.8	4.4	4.1	6.4	6.7	7.2	8.0	5.7	5.9	6.4	4.1	(16)
	PrecMin	12.0	0.0	0.0	0.0	0.0	0.1	0.5	0.1	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	5.1	0.7	1.0	0.8	0.9	1.3	1.4	1.5	1.6	1.4	1.3	1.3	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	60.4	68.1	80.2	86.0	93.1	95.4	96.6	86.4	76.6	64.6	53.4	(19)
		MCWB	48.1	51.1	54.0	61.9	68.6	72.8	71.1	72.0	67.1	64.4	56.5	47.7	(20)
	2%	DB	48.4	55.3	64.1	73.8	82.4	89.4	91.4	91.7	82.2	71.9	59.4	48.4	(21)
		MCWB	44.6	47.8	52.3	58.2	66.1	71.2	71.6	71.2	66.2	61.2	54.0	46.0	(22)
	5%	DB	44.9	50.4	59.2	70.1	78.9	85.4	88.0	87.8	78.4	68.1	56.6	46.0	(23)
		MCWB	42.3	45.3	49.6	56.5	64.5	69.6	69.5	70.2	64.3	59.3	52.0	44.1	(24)
	10%	DB	42.4	46.8	55.5	66.6	75.2	82.0	84.4	84.1	73.8	64.3	53.4	42.9	(25)
		MCWB	40.2	43.2	47.6	54.8	62.5	67.5	68.3	68.6	62.1	57.1	49.7	41.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.0	51.7	56.1	63.8	70.8	75.9	76.1	76.9	69.4	65.4	58.2	49.4	(27)
		MCDB	52.9	58.5	64.8	77.0	83.3	89.5	89.7	91.6	81.6	74.0	62.7	50.9	(28)
	2%	WB	45.6	48.8	53.2	60.2	67.5	72.7	73.4	73.6	67.3	62.5	55.3	46.2	(29)
		MCDB	47.6	53.6	61.1	71.2	79.2	85.6	87.0	86.5	79.1	70.1	59.5	48.2	(30)
	5%	WB	42.5	46.2	50.7	57.8	65.4	70.3	71.7	71.6	65.4	60.2	52.7	43.7	(31)
		MCDB	44.3	49.9	57.2	68.0	76.2	81.8	83.6	83.6	75.5	66.5	56.0	45.0	(32)
	10%	WB	39.7	43.8	48.8	55.7	63.4	68.5	69.9	69.7	63.5	58.1	50.2	41.3	(33)
		MCDB	41.3	46.5	55.0	64.5	73.1	79.1	80.9	81.4	71.8	63.3	52.4	42.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	14.6	18.2	21.1	21.1	21.2	21.6	22.0	19.7	18.7	13.3	10.3	(35)
		MCDBR	15.5	20.4	25.9	27.9	27.6	26.5	27.3	27.6	26.1	24.0	17.8	13.0	(36)
		MCWBR	12.1	14.4	16.3	14.8	13.6	11.4	10.5	11.3	12.1	14.1	12.4	10.6	(37)
	5% WB	MCDBR	14.7	18.6	22.7	25.2	25.3	23.9	23.6	24.5	23.0	22.2	16.3	11.8	(38)
		MCWBR	11.9	13.9	15.2	14.1	13.2	11.3	10.6	11.6	11.7	13.8	11.8	10.1	(39)
(40) Clear-Sky Solar Irradiance	taub		0.330	0.352	0.389	0.438	0.427	0.428	0.441	0.440	0.415	0.383	0.358	0.328	(40)
	taud		2.365	2.317	2.246	2.125	2.216	2.246	2.212	2.223	2.260	2.328	2.352	2.373	(41)
	Ebn at Noon		234	254	261	258	266	266	260	255	249	238	220	219	(42)
	Edh at Noon		23	30	37	46	43	43	43	41	36	29	23	21	(43)
(44) All-Sky Solar Radiation	RadAvg		329	587	983	1450	1750	1900	1861	1669	1174	752	397	249	(44)
	RadStd		39	86	145	139	182	144	133	161	138	81	38	33	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	+1.06	N/A	N/A	N/A	+0.95	+0.97	-216	-315	+175	+86

Nomenclature: See separate page